Does D-Flow help? What the data show

D-Flow can reach high Task 2 success, but the present results do not establish an advantage over tuned QGF. All eight D-Flow conditions have three frozen model-specific choices. Tuned original QGF and FK each scored 40/40 for all three models on fresh confirmation. These are selection/confirmation results, not a new heldout test or a causal comparison of averaging.

Own-best condition	lambda	Seed 11	Seed 23	Seed 37
Regular / raw Q	0	35/40	40/40	37/40
Regular / raw Q	1	28/40	40/40	40/40
Stochastic / raw Q	0	40/40	40/40	39/40
Stochastic / raw Q	1	37/40	40/40	40/40
Regular / averaged Q	0	38/40	40/40	37/40
Regular / averaged Q	1	25/40	40/40	40/40
Stochastic / averaged Q	0	40/40	40/40	40/40
Stochastic / averaged Q	1	35/40	40/40	40/40
Original QGF	-	40/40	40/40	40/40
Original FK	-	40/40	40/40	40/40
Original IPG	-	unqualified	unqualified	unqualified

The weak original F8 BC controls and strong guided settings show useful observed task performance. They do not isolate which guidance mechanism caused improvement: cohorts, hyperparameters and selection differ. Equal numeric kappa across methods is not equal accumulated guidance. The paired starts are not additional trained seeds.

Your rule remains explicit: "Keep the all-33 requirement." Thirty of 33 guided choices are fixed; IPG3 remain unqualified. No new main100 heldout test or matched-kappa curve study has run. Cube has zero policy-performance episodes.

All 24 D-Flow selected tuples

Identity surrogate guidance, full-F8 endpoints, T=1. I is latent updates; K is critic neighbors. The literal3 column is the original three fresh40 outcomes, in frozen order; all72 full alternative tuples are supplied in CSV.

Seed	Mode/Q	L	kappa	eta	I	sigma	K	CONF	Literal3
11	reg/raw	0	0.1	0.1	256	-	-	35/40	35/31/21
11	reg/raw	1	40	0.03	256	-	-	28/40	24/28/0
11	sto/raw	0	5	0.1	128	-	-	40/40	40/24/22
11	sto/raw	1	5	0.1	128	-	-	37/40	37/23/10
11	reg/avg	0	1	0.1	256	0.003	16	38/40	36/38/21
11	reg/avg	1	64	0.03	256	0.5	4	25/40	24/25/24
11	sto/avg	0	5	0.1	128	0.003	16	40/40	40/29/24
11	sto/avg	1	5	0.03	128	0.003	16	35/40	35/26/26
23	reg/raw	0	5	0.1	256	-	-	40/40	40/40/40
23	reg/raw	1	64	0.03	256	-	-	40/40	40/40/40
23	sto/raw	0	80	0.03	256	-	-	40/40	40/40/22
23	sto/raw	1	80	0.01	128	-	-	40/40	40/33/30
23	reg/avg	0	16	0.1	256	0.3	4	40/40	40/40/40
23	reg/avg	1	80	0.03	128	0.5	16	40/40	40/40/40
23	sto/avg	0	64	0.1	128	0.5	4	40/40	40/40/40
23	sto/avg	1	80	0.1	256	0.01	4	40/40	40/40/40
37	reg/raw	0	0.1	0.1	256	-	-	37/40	37/34/25
37	reg/raw	1	5	0.1	128	-	-	40/40	40/22/0
37	sto/raw	0	5	0.1	128	-	-	39/40	39/39/32
37	sto/raw	1	5	0.1	128	-	-	40/40	40/24/4
37	reg/avg	0	0.1	0.1	256	0.003	4	37/40	37/36/23
37	reg/avg	1	5	0.03	256	0.003	4	40/40	40/26/24
37	sto/avg	0	3	0.1	256	0.003	4	40/40	40/34/30
37	sto/avg	1	5	0.03	128	0.003	4	40/40	40/25/25

DF24 original aggregate freeze SHA 19d8af37. Full model/configuration IDs, all tuple values and evidence hashes are in DF24_SELECTED_TUPLES.csv and RESULTS.json. No hidden fourth candidate or retuning is introduced by this report.

Domain: 20 kappas from0 to80; eta .001/.003/.005/.01/.03/.1; I1/2/4/8/16/32/64/128/256; averaged-Q sigma .003/.01/.03/.06/.1/.2/.3/.5; K4/16. The adopted hierarchy is9,240 settings per model, not73,440 Cartesian settings.

QGF, FK and every literal alternative

Selections use only their own frozen fresh40 confirmation outcomes. Their inference integration grids differ: QGF M can differ from trained F8; FK keeps the original actor F8 while using its particle integration M.

Method	Seed	kappa	M	N	noise	Selected
QGF	11	16	64	1	0	40/40
QGF	23	320	8	1	0	40/40
QGF	37	16	16	1	0	40/40
FK	11	80	64	32	0.02	40/40
FK	23	80	32	32	0.02	40/40
FK	37	80	16	32	0.02	40/40

All Task 2 literal fresh40 alternatives

Method	Seed	Order	kappa	M	N	noise	CONF	Chosen
QGF	11	0	20	64	1	0	38/40	
QGF	11	1	16	64	1	0	40/40	yes
QGF	11	2	16	32	1	0	39/40	
QGF	23	0	320	16	1	0	40/40	
QGF	23	1	40	32	1	0	40/40	
QGF	23	2	320	8	1	0	40/40	yes
QGF	37	0	16	32	1	0	39/40	
QGF	37	1	16	16	1	0	40/40	yes
QGF	37	2	12	32	1	0	40/40	
FK	11	0	80	64	32	0.02	40/40	yes
FK	11	1	8	8	16	0	30/40	
FK	11	2	5	64	16	0	37/40	
FK	23	0	80	32	32	0.02	40/40	yes
FK	23	1	5	32	16	0.02	30/40	
FK	23	2	8	16	16	0	29/40	
FK	37	0	80	16	32	0.02	40/40	yes
FK	37	1	5	8	32	0.02	40/40	
FK	37	2	5	64	16	0	39/40	

Complete QGF Task2+Task5 candidate tables contain648 VAL20 cells:27 kappa x4 M x3 models x2 tasks. The original M8 panels are referenced, not charged again as new joint-scan episodes. FK has2160 TRAIN-score rows (720/model); only its declared shortlists have native success outcomes.

Totals: episodes are not score rows

Recorded quantity	Count	Meaning
New native VAL/CONF episodes	30,360	Campaign ledger; not the sum of duplicated metric views
Evaluated active policy action rows	16,185,534	Physical policy controls; model-free controls tracked separately
New main heldout episodes	0	All33 prerequisite remains unmet
DF TRAIN-score rows	27,720	9,240/model; no native success interpretation
DF native candidate metric views	420	348 VAL20 requests +72 fresh40 alternatives
FK TRAIN-score rows / native metric views	2,160 /30	Full720/model scores;21 VAL20 +9 CONF views
QGF full selection cells / literal CONF	648 /18	Both Task2 and Task5, all3 models
Guided choices fixed / required	30 /33	DF24 +QGF3 +FK3; IPG3 unavailable

What "all outputs" includes in this update

The CSV directory includes every recorded DF TRAIN score, all420 DF native candidate metric views, all72 literal alternatives, all648 QGF cells, all18 QGF confirmation alternatives, the FK full TRAIN grid and all30 FK native metric views. All33 statuses and source/model identities are explicit. The source episode CSV index records hashes and roles; it does not create new independent episode credit.

Missing measurements are labelled, not invented: an untested Cartesian DF cell is not a zero; FK TRAIN scores are not720 native rollouts; IPG has no qualified native grid; Cube has no policy utility grid. Opaque traces/checkpoints/raw TRAIN are externally indexed and were not decoded. Historical full assets remain in the unchanged checkpoint archive.

Own-best selection, same-kappa curves and heldout generalization answer different questions. The proposed540 same-kappa points target100 episodes each via20-episode increments, but those curves and their new heldout roles remain unrun.

Harder Cube: training is complete, utility pending

CubeTriple Task3 uses46D observations and25D H5 action chunks (five5D controls), unaugmented local1M TRAIN data, seed11,128x128 networks, fullF8 and T1. This is a limited-data adapted benchmark, not the paper100M-data replication.

Actual stage	Saved result	Scope
Fresh BC50k	50,000 actor/Adam updates; parent91a8	No engineering-update credit
Goal-specific IQL50k	50,000 Q/V updates; fullparent2935	Actor fullstate/RNG frozen; Task3 reward
E1 flat25 callbacks	69 predicates PASS	Interface proof, not sampler or utility
Controlled H5 comparison	31 predicates PASS;40 external controls	Model-free controlled-reset diagnostic
Official reset R0	19 predicates PASS;4 resets/8 internal	0external; equal observations,9 broader state differences
E2a f32 actual	128 cells:120 PASS /8 latent FAIL	No sampler qualification; historical f64 suffix unlaunched
Standalone reference64	3941 eligible FD PASS;2000 AD checks PASS	Separate direct-action probe;59nonsmooth/1000inactive NA
Saved24 reader	8 failing cases retained;296 bad latent coordinates	0action coordinates over original threshold; no clipping-cause proof
I1 first actual attempt	NameError in reference helper during2nd trace	0 baseline/capture B20 calls; no numeric repair evidence

The eight E2a latent failures and895 f32 FD diagnostic flags remain. The standalone64 comparison does not test a full-F8 latent-composed gradient or the failed sampler endpoint. I1 stopped before querying a sampler because a helper module global was not configured; its937 earlier PASS predicates concern earlier gates, not the missing capture. A narrow binding proposal was pending at this frozen snapshot.

TRAIN rewards are per-cube goal count minus3:927,047 at-3;71,460 at-2;1,493 at-1; zero joint-goal rows. There are996,000 full-H5 eligible starts. Coverage is descriptive: neither proof of impossibility nor true IQL Q error. No hidden full-goal BC-success cutoff is imposed.

Unresolved engineering and task evidence

Known-Q high-dimensional fixtures

All596 engineering configurations ran across D8/32/128:585 PASS and11 original-QGF formula-parity FAIL. These are numerical checks on balanced analytic fixtures, not native returns or measured IQL critic error. The largest D32 case387 action difference was1.86511415 (625/640 coordinates), while its kappa0 control passed. Saved named exports do not prove hidden consumer values or a compiler/hardware cause. The large prospective comparison pilot is not adopted.

Original IPG

The materialized-reference B144 gate retains134 PASS /10 FAIL. All144 old production baselines reproduced19 public fields (2736 byte proofs). Old/new failure sets overlap7 cases, with3old-only and3new-only. Small residuals do not replace trajectory checks. Original33 FAIL,15 missing learned cases (including11 oldrho.1 failures), dtype/lowering failures and both ridges remain. No IPG C/native selection is admitted.

Task5 QGF and D-Flow

Seed	Selected QGF kappa	M	Fresh CONF
11	96	16	40/40
23	2	64	36/40
37	80	32	40/40

Task5 tuned QGF scored40/40,36/40,40/40. Four failures on seed23 are real but do not establish a general collapse or averaging repair. Expanded Task5 DF still has four latent-only engineering failures; no expanded native DF utility comparison has run. Custom-reset exposure, zero standard withheld-action exposure and independently tuned task returns are distinct evidence.

The frozen all33 status table includes every unavailable IPG slot. Neither the synthetic known-Q suite nor harder Cube changes that primary requirement. Failures, missing records and historical versions remain preserved.

What Q averaging can and cannot do

Qbar(s,a) = (1/K) sum_j Q(s, clip(a + sigma*epsilon_j))

One flow endpoint receives K nearby critic queries. K is not K flow trajectories or additional physical data. Sigma sets the per-coordinate neighborhood width; its Euclidean radius scales roughly as sigma*sqrt(D). The fixed antithetic bank is held throughout all I latent updates; K4 and K16 need not be nested.

A narrow false peak can be softened if nearby predictions are better. A broad wrong peak can survive. A legitimate narrow good peak can also be blurred. Finite K, clipping, min-head critics and the learned flow mean this is not exact Gaussian smoothing or an exact tilted/Boltzmann sampler. Identity guidance is a surrogate; lambda1 is latent prior pull, not an L1 norm or support detector.

Why performance remains pending on Cube

Actual BC and goal-specific IQL training have completed. The remaining blockers are concrete: qualify the original H5 sampler boundary and replay/reference contract, measure20 original native environments co-resident with the full model during cold compilation, and then adopt the native H5/performance roles and measured budget. R0 two-environment RSS and E1 pure-F8 timing do not prove20-env/4GB fit. A SOURCE-only BC boundary protocol was under review at this snapshot; it is not a performance run.

Official reset action spaces create fresh unowned RNGs. R0 found equal returned observation hashes but nine broader recognized-state differences. Same seed is not a proof of paired simulator starts. Any future performance protocol must disclose its reset law; it cannot silently reuse controlled pinned-Box pairing or reduce the physical cohort to fit.

What would establish a D-Flow benefit

Complete the required qualification and frozen selections, then use the prospectively declared heldout and matched-guidance comparisons with unchanged algorithms, roles and budgets. Current independently selected40-episode panels are useful evidence, but do not establish a causal Qavg effect, true-Q repair, statistical dominance or heldout generalization.

Frozen scope, resources and evidence bundle

Plan365 quantity	Value
Known scientific outer-parent wall, once	45387.763027336 s
Separate old-admission UNKNOWN allowance	600 s; not measured actual wall
Used / initial86,400 s tranche	45987.763027336 s
Remaining / active caps	40412.236972664 s /0
New main100 / Cube policy utility episodes	0 /0
All33 strict user answer	Keep the all-33 requirement

Administrative, child, watchdog and sampler clocks are not added to the scientific outer-parent clock. Model-free native controls and source-implied reset physics are separate ledgers, not policy VAL/CONF episodes. The initial24 worker-hours are a reviewed tranche, not a whole-study completion guarantee. No automatic retry, scope reduction or heldout exception is created by this report.

Included and externally referenced

This compact update contains this PDF/TXT, machine-readable results, complete CSV exports, original CSV/JSON/source evidence copies and SHA manifests. The unchanged35,845,587-byte private plan328 archive (2055 members; SHA59b60fa9...) retains eight exact checkpoint files, six Task2 configurations, derivedP1 data and the original179 asset index. Raw334,519,088-byte Cube TRAIN and the full historical5.868GB workspace archive remain external. This update is not a portable runtime or a whole-workspace transaction.

Frozen input: plan365 SHA7e8666c8... and I1 reconciliation163f129c.... Later source reviews or executions belong to new dated receipts; this report will not be overwritten to incorporate them. No model, array payload, sampler, RNG or environment query was made to generate it. Every final PDF page must pass actual rendered visual QA, followed by root independent content and visual QA, before delivery.

Start with RESULTS.json, ALL33_GUIDED_SELECTION_STATUS_NO_HELDOUT.csv, the full candidate/grid CSVs and SCOPE_AND_MISSING_EVIDENCE_INDEX.csv. Source episode CSVs are indexed by exact path and hash, so duplicated metric views can be distinguished from fresh executions. All failures and unavailable cells remain explicit.